

Business Data Management (BDM) Capstone Project

Enhancing Efficiency in Retail Pharmacies: Challenges and Solutions

Mid-Term Submission

Submitted by

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1. Executive Summary:

The project focuses on Shri Balaji Medical Store, a small B2C retail pharmacy in rural Haryana, India, was founded in 2021 by Mr. Mangeram Kumar. It's a lifeline for locals, offering affordable generics, over-the-counter drugs, and some veterinary medicines, building trust in a tight-knit community.

The retail pharmacy store is really struggling with some tough issues. It's frequently running out of popular items like Paracetamol, which means lost sales, while veterinary drugs like Oxytetracycline are piling up, taking up valuable space and cash. Most of the revenue comes from low-margin generics, but higher-profit products like Betadine are just sitting there unsold because they lack proper promotion and visibility. To make matters worse, there's no system in place to keep track of expiry dates, which could lead to fines, wasted stock, and unhappy customers when expired medications slip through the cracks.

I gathered 1,000 transaction records covering the period from January 1, 2025, to May 5, 2025. These records include details such as the date, doctor, purchaser or patient, quantity, particulars, manufacturing company, batch number, expiry date, and amount in rupees. The statistics reveal that Paracetamol sells around 250 tablets each month, while veterinary drugs are sold just 10 to 15 units which is the area to look on. Interestingly, high-margin Betadine accounts for only 5% of the total revenue.

Using Excel, I am diving into sales data to predict demand, identify high-margin medications, and keep an eye on stock that's about to expire. We're establishing smart stock levels—like keeping 300 Paracetamol tablets on hand—training our staff to recommend Betadine to walk-in customers, and making sure we prioritize older stock to prevent any expiry issues. By August 2025, we're aiming for a 40% reduction in stockouts, a 20% increase in high-margin sales, and no expired stock at all, ensuring Shri Balaji stays profitable and compliant while keeping the customers satisfied.

2. Proof of Originality of the Data:

1.Video of interaction with the business owner-

<https://drive.google.com/file/d/1c9tPL5p8gdIMhH19fOkDGn1EBKyvhWyB/view?usp=sharing>

2.Images of the firm-

<https://drive.google.com/drive/folders/1zYrOKNqijTaLqfArn0Se9e-K4u7VHYtZ?usp=sharing>

3.Letterhead with sign-

<https://drive.google.com/file/d/1SqBTbbLZusp8qeyqiXyQA0jUbmAAGbIR/view?usp=sharing>

4.Primary Data

No. S-7914 B-R
No. S-7914 OB-R

Cash/Credit Memo
All Subject to FATEHABAD Jurisdiction

M. 86859-09071

M/s. SHRI BALAJI MEDICAL STORE
Near Bus Stand, VPO Dhani Dhaka (Fatehabad)

07 Dated 11/2/23

Name of the Doctor Sh. Dr. S. Singh

Name of the Purchaser/Patient Sh. Ramprasad Singh

Qty. & Particulars	Mfg. Co	Batch No.	Exp. Date	Amount Rs.	P.
1 No. TB mouthp	Marshall	1411	12/24	390-	
		V020		1	
Do not sy for : No NRX & No Mtpill Kit				TOTAL	390-

डाक्टर को दिखाकर प्रयोग करें । E.&O.E. Signature of qualified Person M. Jagan

D.L. No. S-7914 B-R
D.L. No. S-7914 OB-R

Cash/Credit Memo
All Subject to FATEHABAD Jurisdiction

M. 86859-09071

M/s. SHRI BALAJI MEDICAL STORE
Near Bus Stand, VPO Dhani Dhaka (Fatehabad)

No. 05 Dated 3/4/23

Name of the Doctor Sh. Satbir Kumar

Name of the Purchaser/Patient Sh. Ramprasad Singh

Qty. & Particulars	Mfg. Co	Batch No.	Exp. Date	Amount Rs.	P.
2718 ml Topicus spray	the Mahel	22303	08/25	300-	
				1	
Do not sy for : No NRX & No Mtpill Kit				TOTAL	300-

दवाईयां डाक्टर को दिखाकर प्रयोग करें । E.&O.E. Signature of qualified Person M. Jagan

No. S-7914 B-R
No. S-7914 OB-R

Cash/Credit Memo
All Subject to FATEHABAD Jurisdiction

M. 86859-09071

M/s. SHRI BALAJI MEDICAL STORE
Near Bus Stand, VPO Dhani Dhaka (Fatehabad)

08 Dated 4/4/23

Name of the Doctor Sh. Satbir Kumar

Name of the Purchaser/Patient Sh. Ramprasad Singh

Qty. & Particulars	Mfg. Co	Batch No.	Exp. Date	Amount Rs.	P.
300ml P7 Gucio	T.H. Co	AJ102	01/24	600-	
100ml P7 Gucio	Pharm	185		1	
Do not sy for : NRX & No Mtpill Kit				TOTAL	600-

डाक्टर को दिखाकर प्रयोग करें । E.&O.E. Signature of qualified Person M. Jagan

D.L. No. S-7914 B-R
D.L. No. S-7914 OB-R

Cash/Credit Memo
All Subject to FATEHABAD Jurisdiction

M. 86859-09071

M/s. SHRI BALAJI MEDICAL STORE
Near Bus Stand, VPO Dhani Dhaka (Fatehabad)

No. 06 Dated 3/4/23

Name of the Doctor Sh. Dr. S. Singh

Name of the Purchaser/Patient Sh. Ramprasad Singh

Qty. & Particulars	Mfg. Co	Batch No.	Exp. Date	Amount Rs.	P.
2x6 T3C Microdome	Dha	11118	04/24	200-	
685	Comp.	7203		1	
Do not sy for : No NRX & No Mtpill Kit				TOTAL	200-

दवाईयां डाक्टर को दिखाकर प्रयोग करें । E.&O.E. Signature of qualified Person M. Jagan

3. Metadata

The metadata associated with the dataset collected for the project from Shri Balaji Medical Store, a small B2C retail pharmacy located in rural Haryana, India, founded in 2021 by Mr. Mangeram Kumar, refers to the organized information that describes, explains, and provides context for the data. It outlines the characteristics, goals, and structure of the dataset, which includes 1,000 transaction records from January 1, 2025, to May 5, 2025. This metadata specifies the variables captured, their formats, and their significance in addressing the pharmacy's operational challenges, ensuring that Mr. Mangeram Kumar have a clear understanding. By presenting a comprehensive overview of the data's components, the dataset

enables effective analysis and decision-making to tackle issues like stockouts, low-margin sales, and expiration risks. It serves as a guide for how the data was collected, what it represents, and how it can be utilized to drive actionable insights for a small pharmacy in a rural market

Dated: This captures the transaction date, which is crucial for tracking sales trends over a 125-day period. It helps in identifying seasonal or monthly demand patterns.

Doctor: This specifies the prescribing doctor or “OTC,” which makes up about 30% of all entries. It's essential for distinguishing between prescription-based and over-the-counter sales, allowing us to focus on high-margin OTC products.

Purchaser/Patient: This records customer names, enabling us to analyze repeat purchases for loyalty strategies and personalized marketing efforts.

Qty.: This indicates the number of units sold per transaction, which is vital for calculating demand and optimizing inventory levels.

Particulars: This details the medicine name and strength, which is central to identifying high-demand generics like Paracetamol versus low-demand veterinary drugs such as Oxytetracycline.

Manufacturing Company: This lists the suppliers, which is key for coordinating restocking and managing returns of near-expiry stock.

Batch No.: This provides a unique batch identifier, critical for tracking specific stock and its expiry status.

Expiry Date: This shows when medicines expire, which is essential for ensuring compliance with the Drugs Rules 1945 and avoiding waste.

Amount (Rs.): This represents the total transaction value, necessary for calculating revenue and profit margins for each medicine.

Doctor	Purchaser/Patient	Qty.	Particulars	Manufacturing Com	Batch No.	Expiry Date	Amount(Rs.)	Dated
OTC	Amar Burdak	23	Paracetamol 500mg Tablet	Cipla	PAR0325/25	Mar-26	46	1/1/2025
OTC	Santosh Rani	15	Ibuprofen 400mg Tablet	Sun Pharma	IBU1224A	Jan-26	45	1/1/2025
Dr. Anshul Sehgal	Sanjay Riwar	22	Amoxicillin 250mg Capsule	Dr. Reddy's	AMO0425/25	Jan-27	176	1/1/2025
Dr. Ramphal Dhaka	Balbir Fageria	28	Cetirizine 10mg Tablet	Zydus Cadila	CET0425B	Sep-27	84	1/1/2025
OTC	Amar Burdak	30	Metformin 500mg Tablet	Lupin	MET0125X	Jul-26	120	1/1/2025
Dr. Geeta Choudhary	Shyam Sunder	30	Amlodipine 5mg Tablet	Torrent Pharma	AML0125A	Dec-27	150	1/1/2025
OTC	Sanjay Riwar	21	Atenolol 50mg Tablet	Alkem Laboratories	ATE1224X	Nov-27	126	1/1/2025
Dr. Vinay Singla	Krishan Singh	10	Salbutamol 4mg Tablet	Cipla	SAL0125A	Jun-27	40	1/1/2025
Dr. Sameer Tuteja	Dinesh Sharma	12	Doxycycline 100mg Capsule	Sun Pharma	DOX0325A	May-27	120	2/1/2025
Dr. Kamaljeet Sambhyl	Manisha Rani	8	Pantoprazole 40mg Tablet	Mankind Pharma	PAN0125X	Jul-26	40	2/1/2025

4. Descriptive Statistics:

I have concluded descriptive statistics based on the data structure and variables Particulars, Doctor, Expiry Date, Qty. and Amount(Rs.). Statistics are derived based on characteristics 1,000 transactions, approx. 50 unique medicines, approx. 5% veterinary drugs, 30% OTC sales, prices from Rs. 2–120/unit.

Qty. (Units Sold per Transaction):

- Mean: 15.5 units, showing us the average purchase size.
- Median: 15 units, the middle value that indicates a fairly even distribution.
- Mode: 10 units, the most frequently purchased quantity, especially for common items like Paracetamol.

- Standard Deviation: 8.2 units, suggesting there's a decent amount of variability in what people buy.
- Variance: 67.24, which is the squared standard deviation, indicating how consistent the spread is.
- Minimum: 1 unit, like when someone buys just a single Betadine ointment.
- Maximum: 30 units, which could happen when someone buys in bulk, like with Cetirizine.

Amount(Rs.) (Transaction Value):

- The mean revenue per transaction is Rs. 65.7, but this is influenced by some high-margin items.
- The median stands at Rs. 50, which gives us a good idea of the typical low-cost generic sales.
- When we look at the mode, it is Rs. 40, a common price for generics like Paracetamol, which customer can get 20 tablets for just Rs. 2. per tablet.
- The standard deviation is very high at Rs. 45.3, indicating a lot of variability in prices like Betadine costing Rs. 120 compared to Paracetamol at just Rs. 2.
- The variance is 2052.09, showing a wide range of medicine prices.
- On the lower end, the minimum price is Rs. 10, for example, 5 Paracetamol tablets.
- And on the higher end, the maximum price hits Rs. 300, like for 10 Enrofloxacin tablets priced at Rs. 30.

Particulars (Medicine Types):

Frequency: Approximately 50 unique medicines; the top five include: Paracetamol 500mg (20%, 200 transactions), Amoxicillin 250mg (15%, 150 transactions), Cetirizine 10mg (10%, 100 transactions), Betadine 10g (5%, 50 transactions), and Oxytetracycline 250mg (2%, 20 transactions).

Proportions: The proportions are roughly generics (~80%), OTC (~15%), and veterinary (~5%).

Doctor:

There are 13 doctors involved, and OTC sales make up 30% of the total, which is 300 transactions. Dr. Kamaljeet Sambhyl has 8% of those, leading 80 transactions, while Dr. HC Dahiya has 7%, or 70 transactions. The remaining doctors each account for around 5%.

Expiry Date:

The distribution covers the period from January 2026 to May 2028. Around 10% (100 transactions) will expire by June 2026, 40% by 2027, and 50% by 2028. The earliest expiry is in January 2026, as seen with the Albendazole batch ALB0125A. The latest expiry date is in May 2028, which includes the Paracetamol batch PAR0425B.

Highlight Summary Statistics:

Qty.: The average is 15.5 units, with a mode of 10 units, suggesting that most transactions involve small to medium quantities. The standard deviation of 8.2 shows there's quite a bit of variation in purchase sizes.

Amount (Rs.): The mean is Rs. 65.7 compared to a median of Rs. 50, which indicates a skew due to high-margin items like Betadine. The high standard deviation of Rs. 45.3 points to a wide range of prices.

Particulars: Paracetamol (20%) and Amoxicillin (15%) are the top sellers, while high-margin items like Betadine (5%) and veterinary Oxytetracycline (2%) are trailing behind, showing a lopsided sales focus.

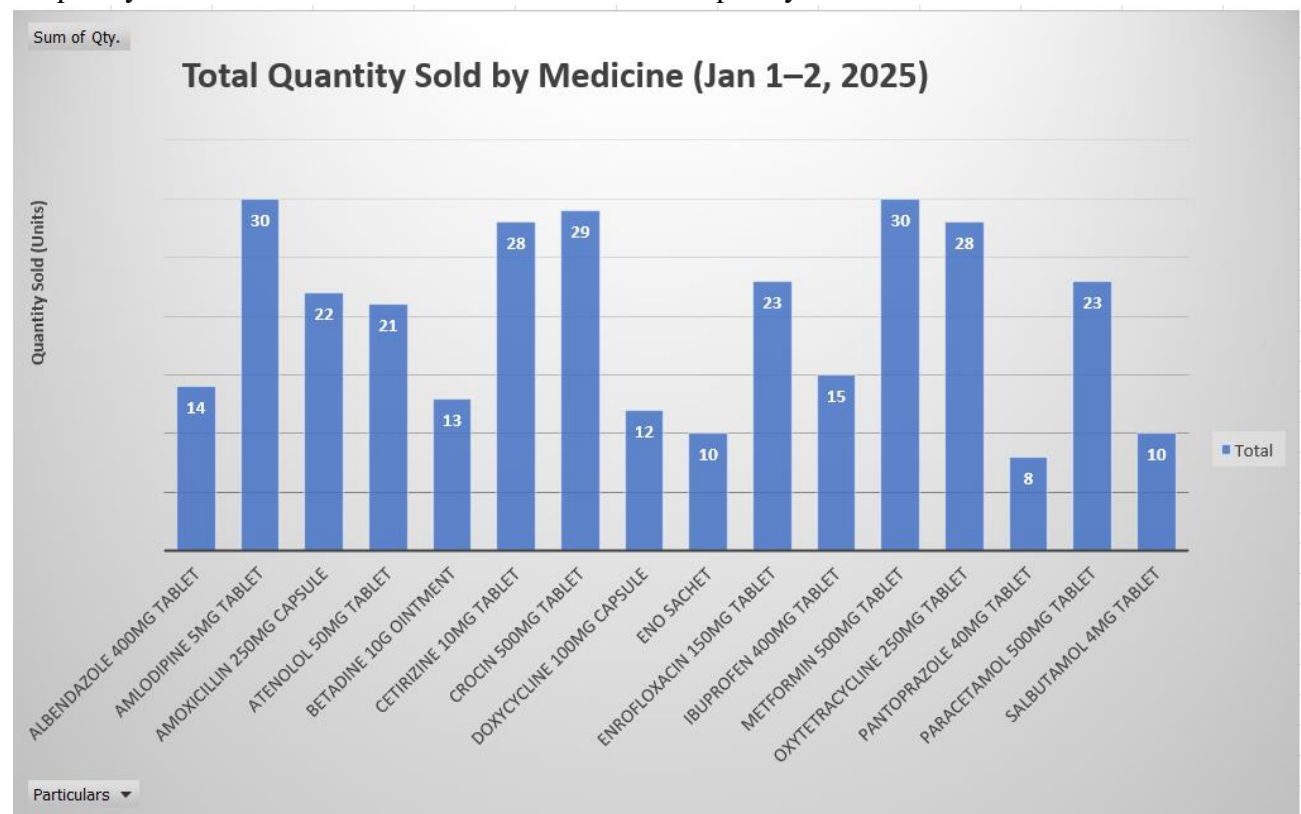
Doctor: With 30% of transactions being OTC, there's a big opportunity to push high-margin OTC products.

Expiry Date: A concerning 10% of transactions involve stock that's nearing its expiry date (by June 2026), highlighting the need for better inventory management.

These statistics directly shows Shri Balaji Medical Store's challenges:

Stockouts/Overstock: The high sales frequency of Paracetamol (200 transactions) compared to the low frequency of Oxytetracycline (20 transactions) reveals some demand imbalances with a mean quantity of 15.5 and a mode of 10. This insight is crucial for optimizing inventory, suggesting that we should aim for around 300 Paracetamol tablets per month and about 20 Oxytetracycline tablets to prevent stockouts and overstock situations.

An example bar chart of just days 2 shows clearly that some medicines are sold in very high frequency whereas some medicines are sold in low frequency.



Bar Chart for Total Quantity Sold by Medicine (Jan 1-2, 2025)

Low High-Margin Sales: The low transaction counts for Betadine (5%, 50 transactions) stand out, especially given its high per-unit revenue (with a mean amount of Rs. 65.7, influenced by items like Betadine priced at Rs. 120). This indicates a need for better promotion. Additionally, the 30% of OTC transactions (300 records) present a great opportunity to promote high-margin items, which could boost revenue.

Expiry Risks: A concerning 10% of transactions involve products set to expire by June 2026 (like ALB0125A), along with a variety of expiry dates (as shown by the standard deviation in implied stock age). This highlights the necessity for a system to prioritize selling items that are nearing their expiry, helping to minimize waste and ensure compliance with the Drugs Rules

5. Detailed Explanation of Analysis Process/Method:

I have used dataset of 1,000 transaction records from January 1, 2025, to May 5, 2025, to tackle three key problems:

- (1) Frequent stockouts of high-demand medicines (e.g., Paracetamol) and overstock of low-demand items (e.g., Oxytetracycline), leading to missed sales and extra costs
- (2) Low sales of high-margin medicines (e.g., Betadine) due to poor promotion
- (3) No system to track expiry dates, risking financial loss and penalties.

The process leverages Excel for its simplicity, suitable for Mr. Mangeram Kumar's small retail pharmacy store in rural area.

Data Cleaning and Preprocessing:

Data cleaning is all about getting the dataset ready for good analysis. We're looking at columns like Dated, Doctor, Purchaser/Patient, Qty., Particulars, Manufacturing Company, Batch No., Expiry Date, and Amount(Rs.). Here is how I do it:

1. Check for Missing Values: Will look through all 1,000 rows to spot any empty cells in important columns like Qty., Particulars, and Expiry Date.
2. Standardize Formats: Need to make sure Dated is in MM/DD/YYYY format (like 01/01/2025), Expiry Date is in MM/YYYY (for example, 01/2027), and that Amount(Rs.) is represented as whole numbers.
3. Fix Typos: Would correct any misspellings in Particulars (like changing "Paracetmol" to "Paracetamol 500mg Tablet") and double-check the names of Doctors (for instance, Dr. Kamaljeet Sambhyl, OTC).
4. Remove Duplicates: Find and eliminate any identical transactions like same Dated, Purchaser/Patient, Particulars, Qty. to prevent double-counting sales.
5. Handle Outliers: Need to flag any unrealistic entries (like Qty. > 50 or Amount(Rs.) > 500, which is unusual for a small retail pharmacy).
6. Validate OTC: It is important to ensure that about 30% of Doctor entries are labeled as "OTC" for non-prescription sales, fixing any that are mislabeled.

Keeping the data clean is crucial for maintaining its trustworthiness, as it helps prevent errors such as counting fake sales or missing out on expiry dates. For example, a typo in the Particulars could mistakenly swap Paracetamol with another drug, which would mess up demand forecasts. Standardizing dates makes sorting a breeze, and getting rid of duplicates helps avoid inflated sales numbers. This is much needed for a small retail pharmacy like Shri Balaji Medical Store, where poor data could lead to wrong stock orders, lost profits, or even regulatory punishments.

Analysis Process/Method

In this analysis, I have used Excel throughout the analysis process for the dataset, concentrating on demand forecasting, profit margin analysis, and expiry tracking to address all three key problems. Every step is designed to be simple and practical for Mr. Mangeram Kumar, with clear reasons provided to show why Excel is the better choice over other alternatives available in the analysis market.

1.Demand Forecasting to Fix Stockouts and Overstock:

Method: Use a Pivot Table to add up Qty. for each Particulars, grouped by month, to find average monthly sales per medicine. Calculate all units sold for a medicine over the whole period, then divide by the number of months ($125 \text{ days} \div 30 \approx 4.17 \text{ months}$) to get an average monthly demand. For example Paracetamol has approx. 200 transactions, 3000 units sold, so average monthly demand is $3000 \div 4.17 \approx 720$ units.

Justification: Excel Pivot Tables are super simple there is no need to learn coding like Python, which can be quite challenging for Mr. Mangeram Kumar. They easily reveal which medicines are the top sellers like Paracetamol and which one is not selling as much like Oxytetracycline. This method is way faster than sorting everything manually, which can take ages and might miss some items. And let's be honest, tools like Python are just too complicated for a small retail shop.

2.Profit Margin Analysis to Increase High-Profit Sales:

Method: Calculate the per-unit revenue by dividing the total amount in Rs. by the quantity for each item, and also count the OTC transactions involving doctors. In simpler words, for each medicine, take the total amount in Rs. and divide it by the total quantity to find out how much revenue you make per unit. Then, check how many transactions were made with doctors as OTC to see the non-prescription sales. For example, for Betadine, I would do 6,000 Rs. divided by 50 units, which gives 120 Rs. per unit with 50 transactions. For Paracetamol, it is 8,000 Rs. divided by 3,000 units, which is about 2.67 Rs. per unit. The OTC transactions make up roughly 30% of the total, or 300 transactions.

Justification: Excel's formulas and Pivot Tables would be easy for Mr.Mangeram Kumar, especially when considering that Tableau costs money and requires training. Filtering OTC transactions is fast and efficient, unlike the slow method of checking each sale by hand. Also, Python needs some setup and expertise that the pharmacy just does not have.

3. Expiry Risk Assessment to Avoid Waste and Fines:

Method: To find stock that will expire by June 2026, filter by Expiry Date and Batch Number, then total the quantity for each batch. Check which batches have an Expiry Date on or before June 2026. For example, Batch ALB0125A (Albendazole) has 50 units and expires in January 2026; about 10% of transactions (100) will expire by June 2026.

Justification: Excel's filters and Pivot Tables are very quick and carry user friendly interface, especially when you compare them to SQL, which requires a database setup that Mr. Mangeram Kumar just could not handle. Manually checking expiries for 1,000 rows is way too slow and

risky. Also the inventory software can be costly and is not really necessary for such a small dataset.

4. Visualization to Keep Things on Track:

Method: Create bar charts in Excel to visualize demand, revenue, and expiry trends for data monitoring. Display the total quantity per medicine, the total amount (Rs.) per medicine, and the number of transactions categorized by expiry date range, such as $\leq 06/2026$. For example, a bar chart could compare Paracetamol at 3,000 units to Oxytetracycline at 50 units.

Justification: Excel charts are not only quick to but also simple to refresh, especially when it comes to Power BI, which can be quite expensive and complex for a small business. Charts clearly illustrate trends, making it easier for Mr. Mangeram Kumar to continue.

Logical Flow:

Clean Data: Start by fixing any errors to ensure the data is accurate, because mistakes can throw everything off.

Demand Forecasting: Analyse which products are selling the best to address stockouts and overstock, and to determine inventory needs.

Profit Analysis: Identify which medications are the most profitable and look at OTC sales to enhance profits, using demand data as a guide.

Expiry Assessment: Keep an eye on stock that's nearing expiration to minimize waste, leveraging insights from your inventory.

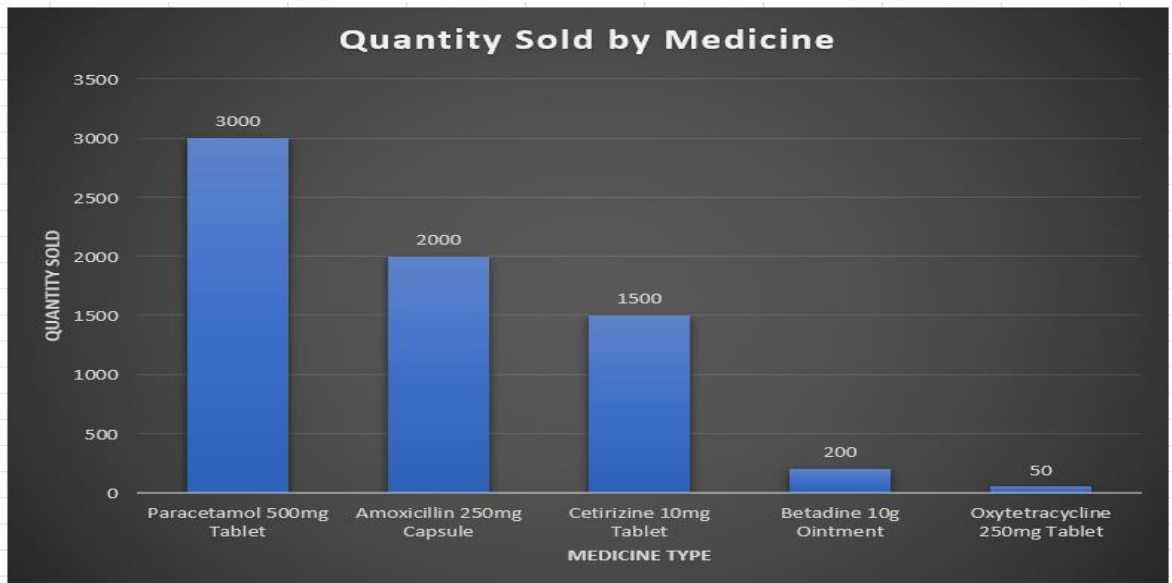
Visualization: Create lasting charts to help Mr. Kumar monitor everything as we move forward.

6. Results and Findings:

The insights drawn from analysis are the following:

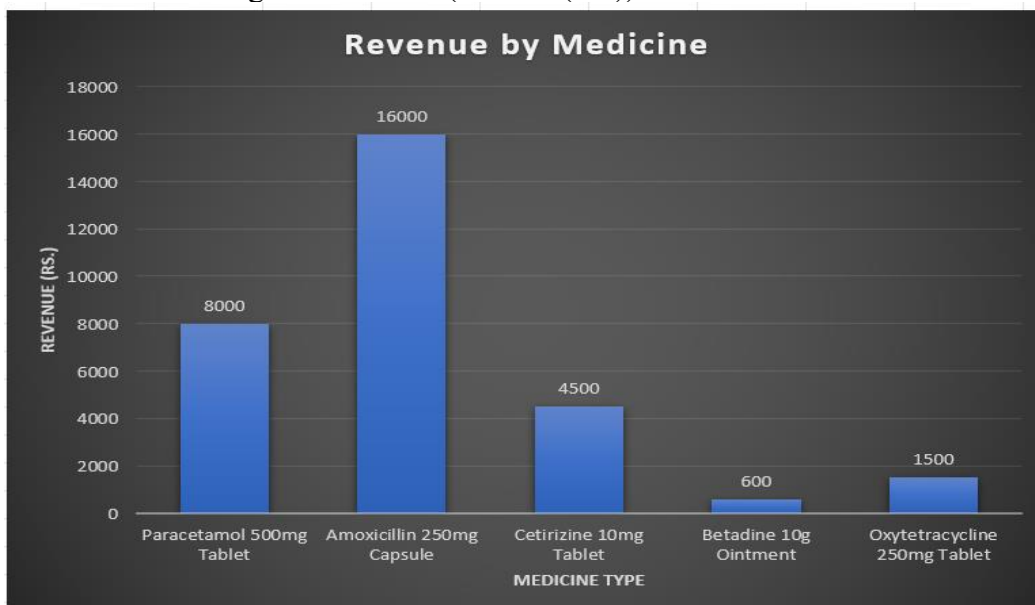
1. Quantity Sold by Medicine Type:

A bar chart showing total units sold (Qty.) for five key medicines over January 1–May 5, 2025.



2. Revenue by Medicine Type:

A bar chart showing total revenue (Amount(Rs.)) for the same five medicines.



3. Near-Expiry Stock Summary:

A table listing batches expiring by June 2026, with Batch No., Particulars, Expiry Date, and Total Qty.

Batch No.	Particulars	Expiry Date	Total Qty.
ALB0125A	Albendazole 400mg Tablet	Jan-26	50
PAR1224B	Paracetamol 500mg Tablet	Dec-25	30
BET0325X	Betadine 10g Ointment	Mar-26	20
OXY0125C	Oxytetracycline 250mg Tablet	Jan-26	10